

IF3170 Inteligensi Artifisial

Institut Teknologi Bandung

Decision Tree

Semester I TA 2026/2027



Variable (Attribute) Types

Nur ULFA Maulidevi

KK IF - Teknik Informatika- STEI ITB

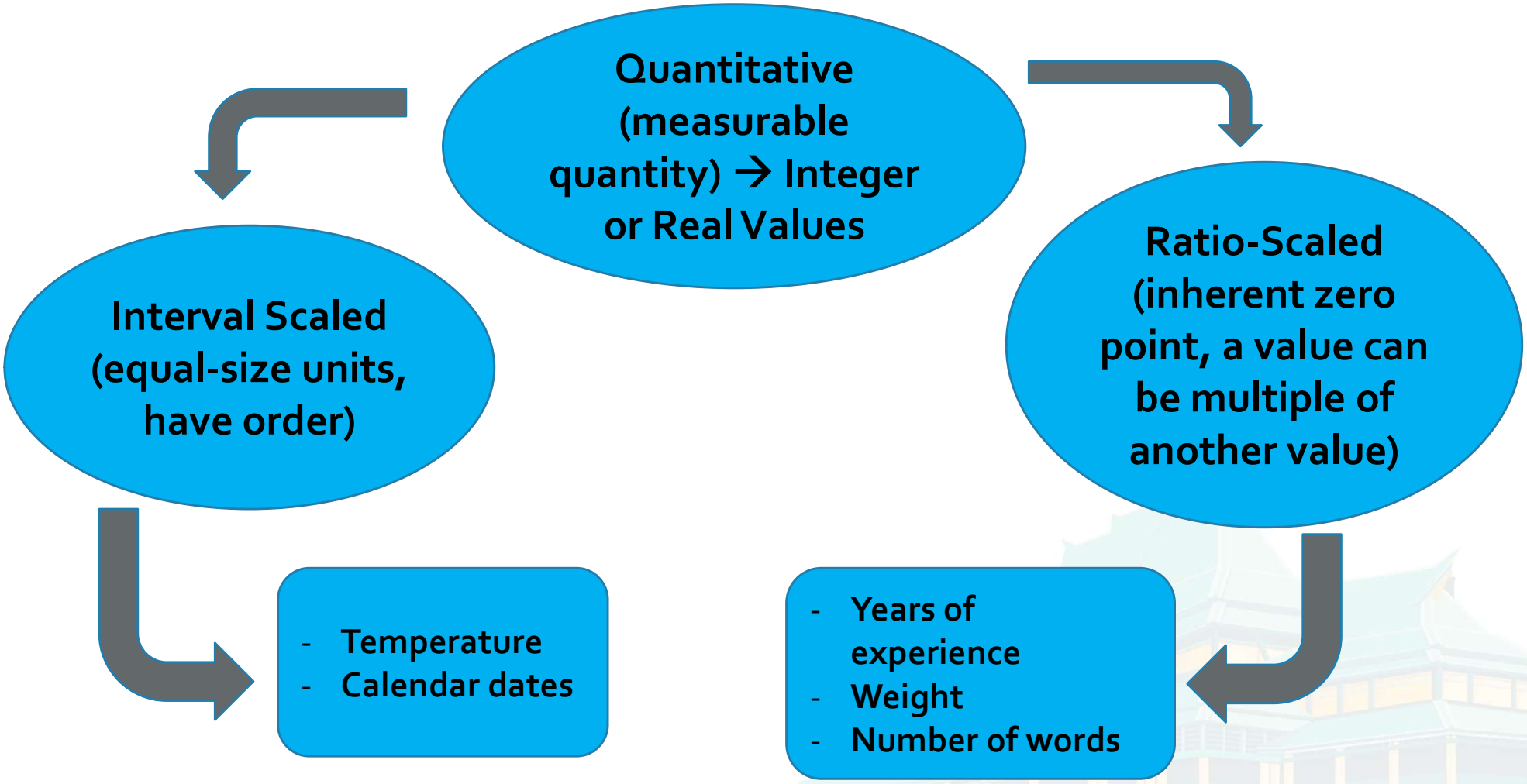
Modul : Decision Tree Learning (DTL)

Variable (Attribute) Types

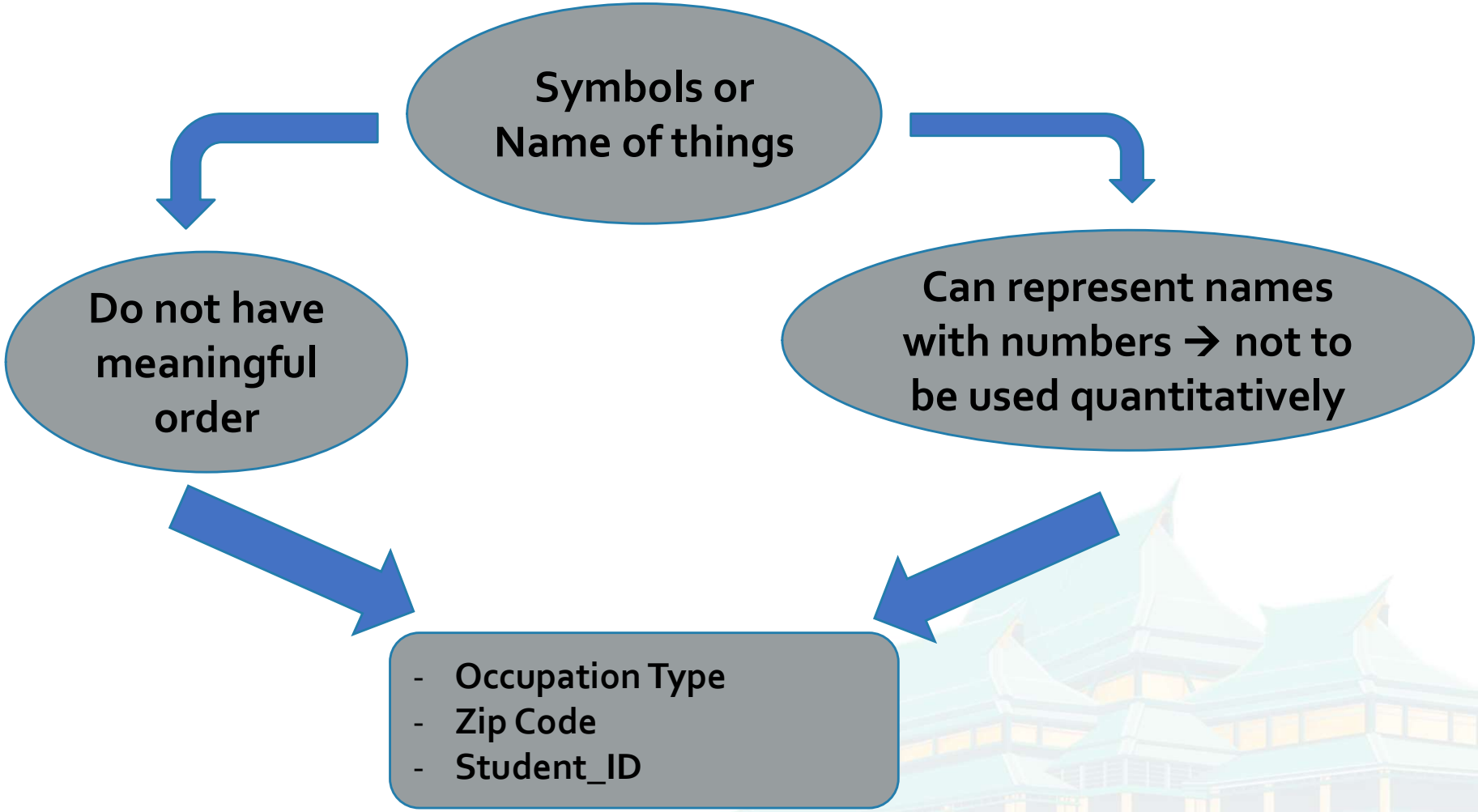
Source: DataMining Concepts and Techniques by Jiawei Han, Micheline Kamber, Jian Pei

Pembelajaran Mesin
(Machine Learning)

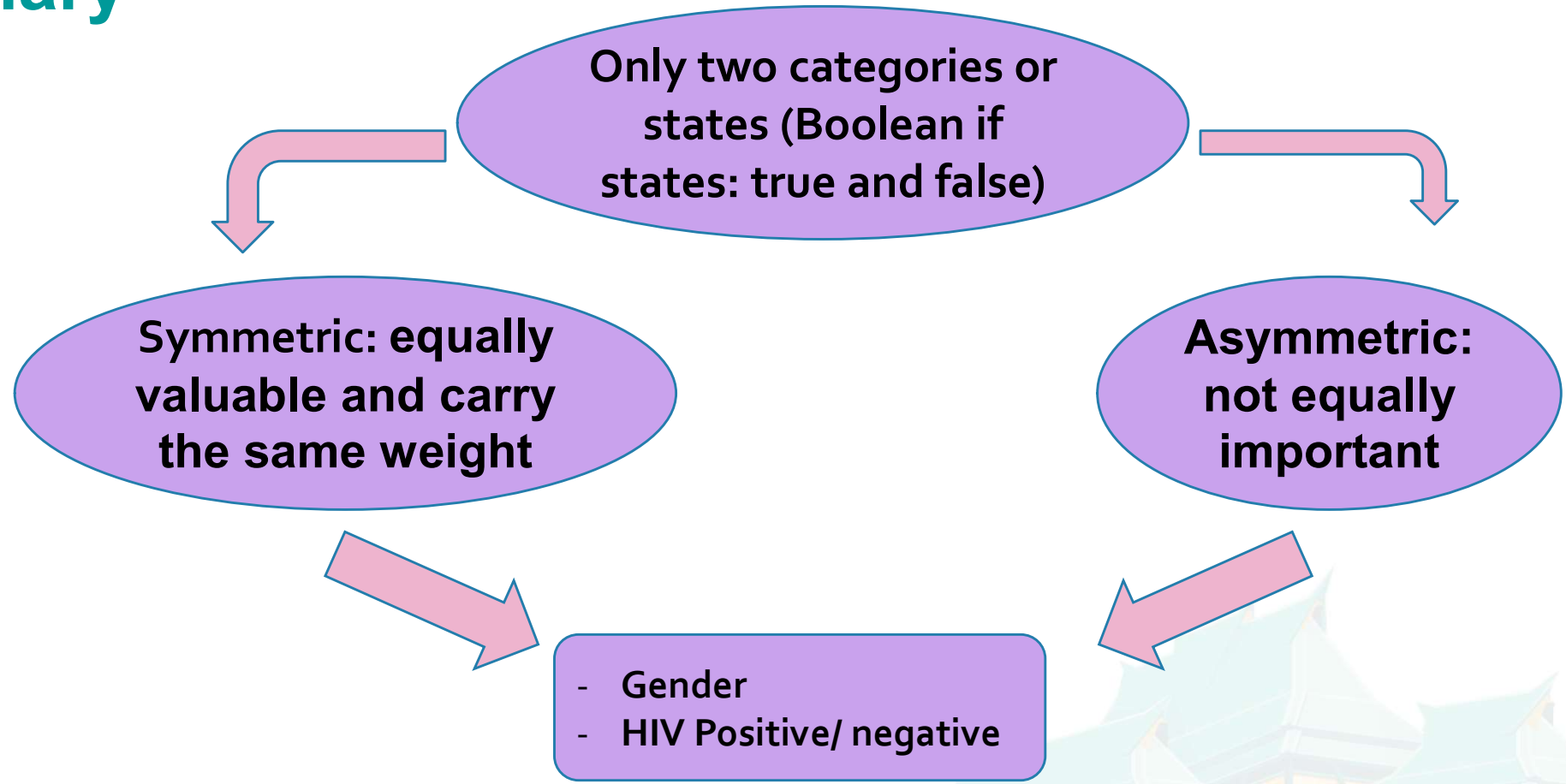
Numeric


Quantitative (measurable quantity) → Integer or Real Values
Interval Scaled (equal-size units, have order)
Ratio-Scaled (inherent zero point, a value can be multiple of another value)
- Temperature
- Calendar dates
- Years of experience
- Weight
- Number of words

Nominal/ Categorical

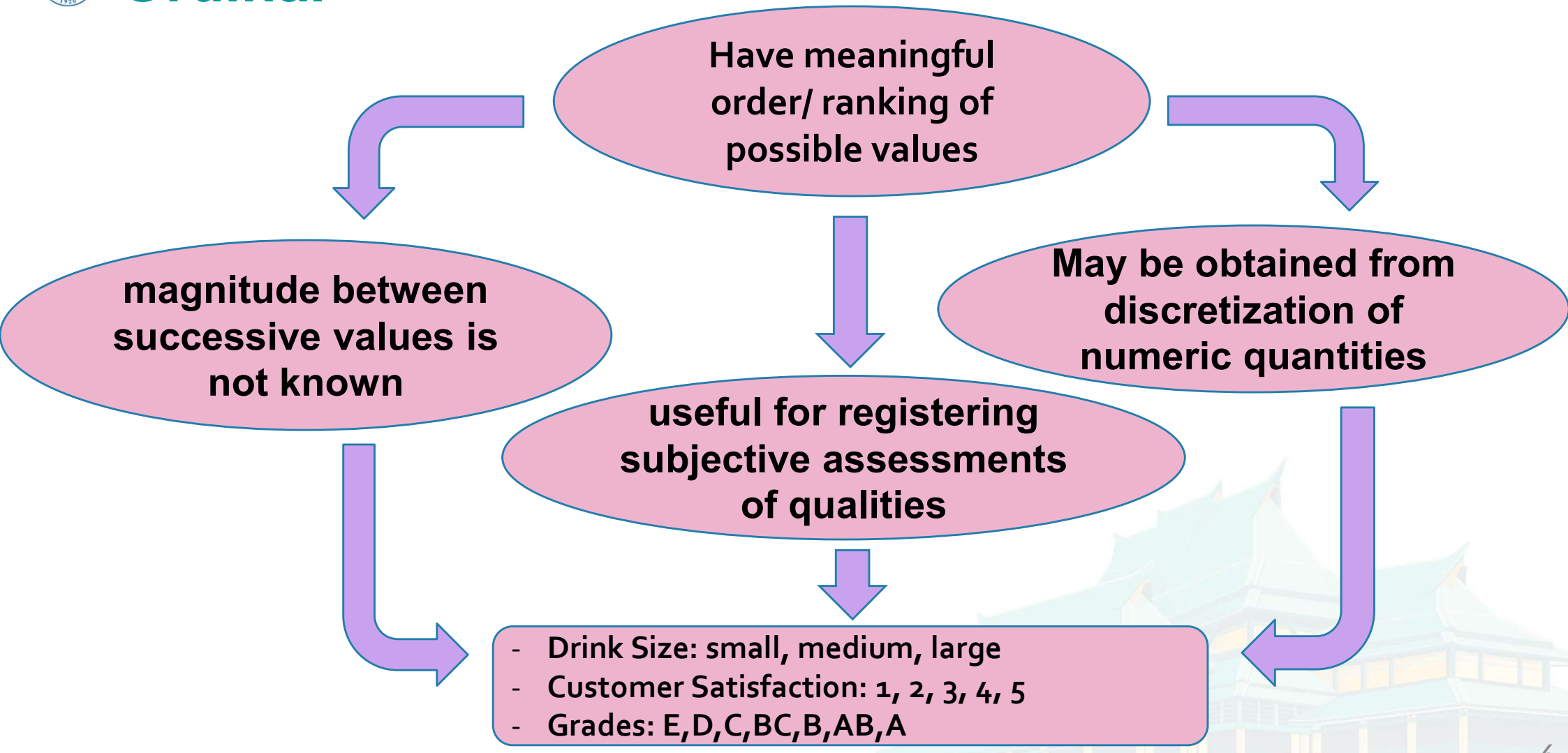


Binary




```
1 red,    green,  blue
2 1,      0,      0
3 0,      1,      0
4 0,      0,      1
```

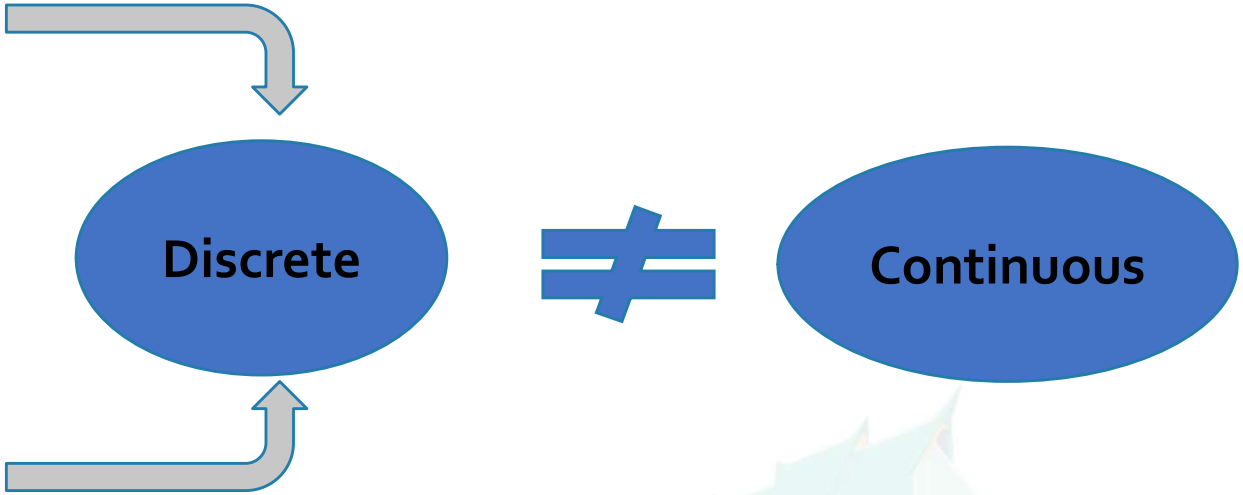
Ordinal


Have meaningful order/ ranking of possible values
magnitude between successive values is not known
May be obtained from discretization of numeric quantities
useful for registering subjective assessments of qualities
- Drink Size: small, medium, large
- Customer Satisfaction: 1, 2, 3, 4, 5
- Grades: E,D,C,BC,B,AB,A

Discrete vs Continuous

has a finite set of values:
Drink size, Age, Medical test,

has a countably infinite set of values:
Customer ID, Zip code



What, Why, and When

Modul : Decision Tree Learning (DTL)

What, Why, and When

Source: Machine Learning, Tom Mitchell

Nur ULFA Maulidevi

KK IF - Teknik Informatika- STEI ITB

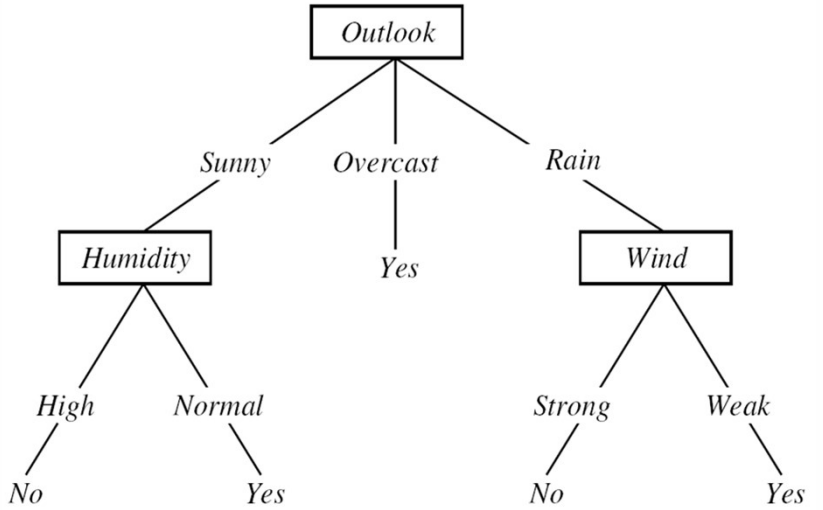
Pembelajaran Mesin
(Machine Learning)

WHAT is DTL

Method for approximating discrete-valued target functions

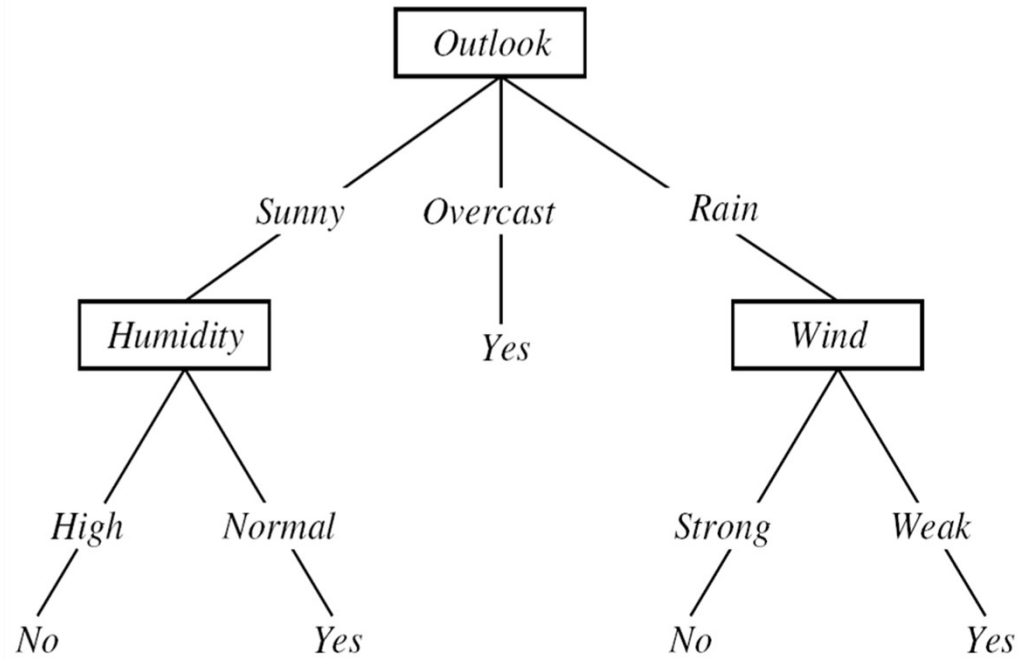
Represented by Decision Tree

Can be represented as set of if-then rules



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

DTL Representation



- Each internal node represents test of an attributes
- Branch descending from the node corresponds to one possible value
- Leaf/ terminal nodes represents classification result

Represent a disjunction of conjunctions of constraints on the attribute values of instances

(outlook=sunny ∧ humidity=normal)
∨ (outlook=overcast)
∨ (outlook=rain ∧ wind=weak)

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

WHY DTL

Method for approximating discrete-valued target functions

Represented by Decision Tree or If-Then Rules

Robust to Noisy Data

Capable of learning disjunctive expression

- Popular of inductive inference algorithms
- Have been successfully applied to a broad range of tasks: learning to diagnose medical cases, learning to assess credit risk of loan applicants

When we use DTL → Appropriate Problems for DTL

Instances are represented by attribute-value pairs → For continuous attribute: discretization

The target function has discrete output values

Disjunctive descriptions may be required

The training data may contain errors

The training data may contain missing attribute values

Basic DTL Algorithm (ID3)

Nur ULFA Maulidevi

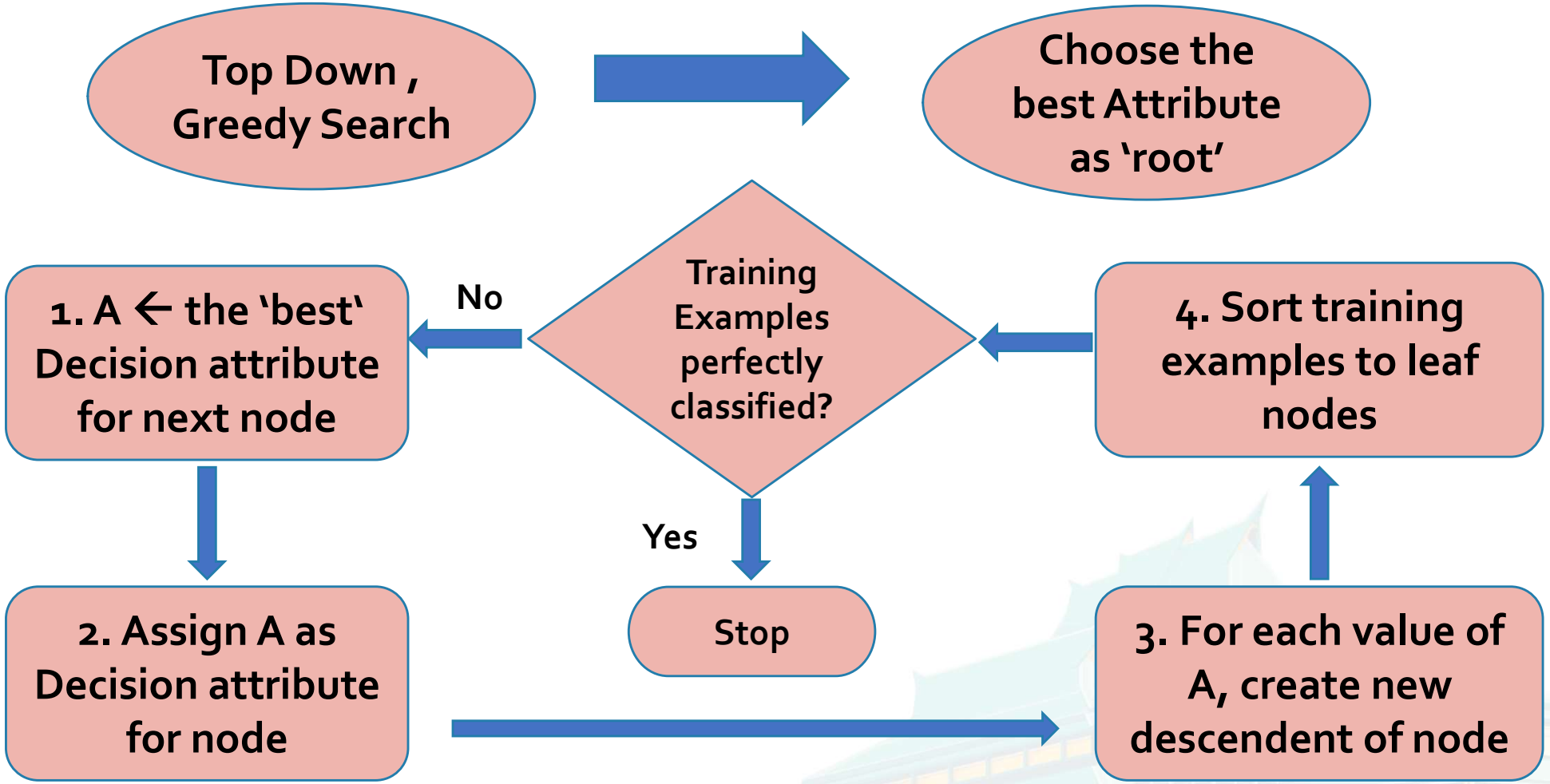
KK IF - Teknik Informatika- STEI ITB

Basic DTL Algorithm (ID3)

Source: Machine Learning, Tom Mitchell

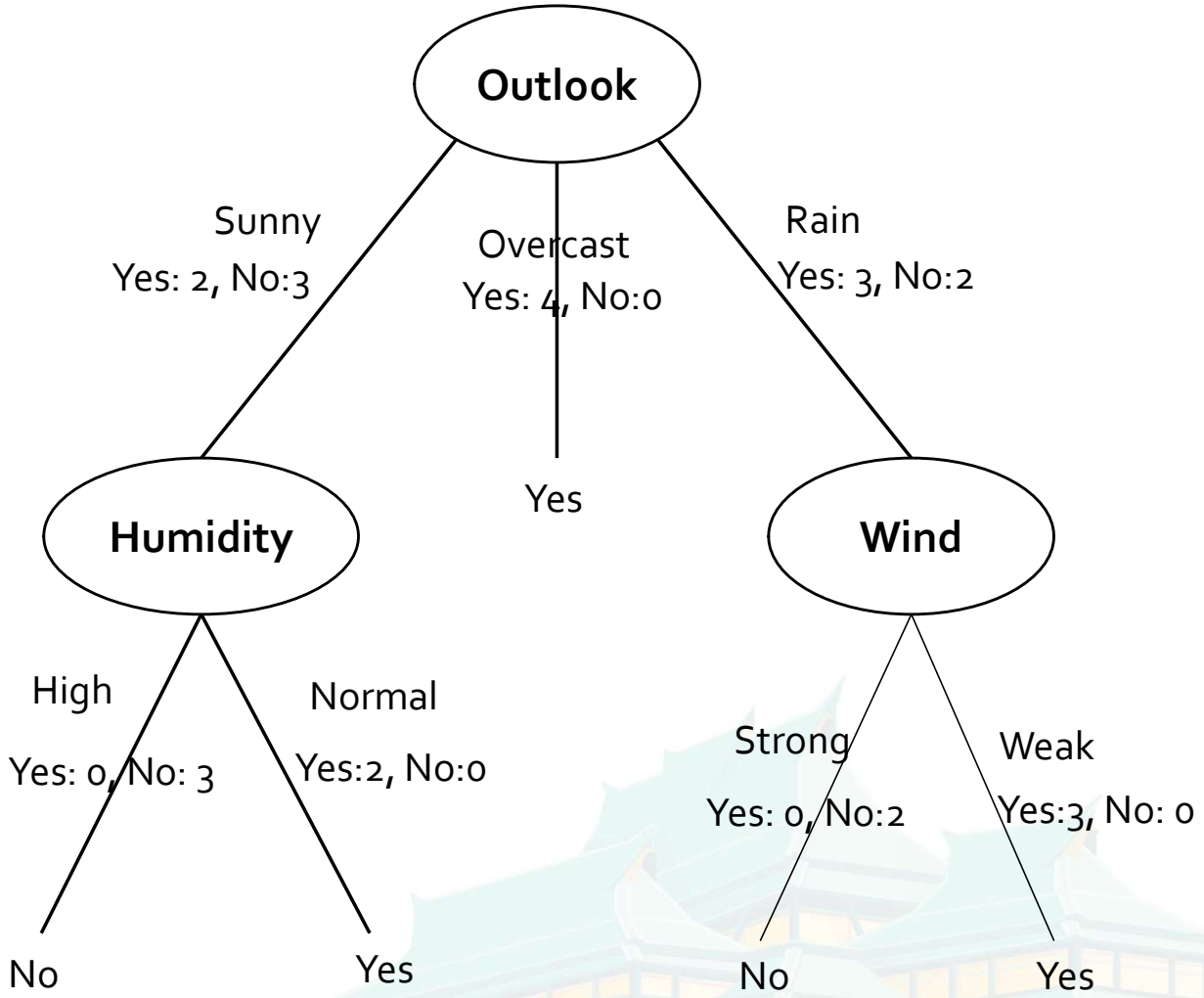
Pembelajaran Mesin
(Machine Learning)

ID3 Algorithm



Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Decision Tree Learning (Russel & Norvig, 2021)

function DECISION-TREE-LEARNING($examples$, $attributes$, $parent_examples$) **returns** a tree

if $examples$ is empty **then return** PLURALITY-VALUE($parent_examples$)
else if all $examples$ have the same classification **then return** the classification
else if $attributes$ is empty **then return** PLURALITY-VALUE($examples$)
else
  $A \leftarrow \text{argmax}_{a \in attributes}$ IMPORTANCE(a, $examples$)
  $tree \leftarrow$ a new decision tree with root test A
  for each value v_k of A **do**
    $exs \leftarrow \{e : e \in examples$ **and** $e.A = v_k\}$
    $subtree \leftarrow$ DECISION-TREE-LEARNING(exs, $attributes - A$, $examples$)
    add a branch to $tree$ with label $(A = v_k)$ and subtree $subtree$
  return $tree$

Information gain
Gain ratio

|A|=2: binary tree
|A| min 2: n-ary tree

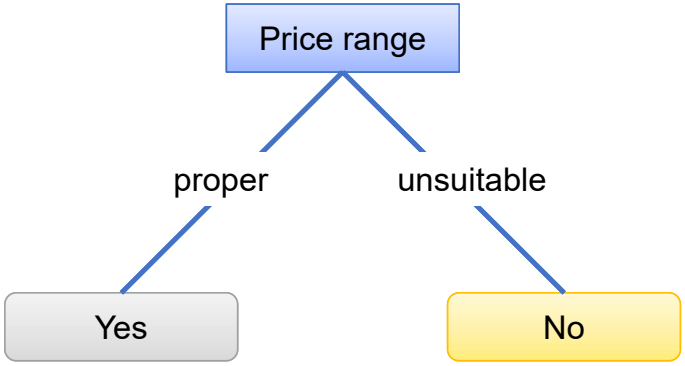
The function PLURALITY-VALUE selects the most common output value among a set of examples, breaking ties randomly.

Example 1: Dining out data

Restaurants	Food quality	Service speed	Price range	Dine out?
P1	Good	Strong	Proper	Yes
P2	Good	Moderate	Proper	Yes
P3	Good	Moderate	Unsuitable	No
P4	Good	Weak	Unsuitable	No
P5	Average	Strong	Proper	Yes
P6	Average	Moderate	Proper	Yes
P7	Average	Moderate	Unsuitable	No
P8	Average	Weak	Unsuitable	No
P9	Poor	Strong	Proper	Yes
P10	Poor	Moderate	Proper	Yes
P11	Poor	Moderate	Unsuitable	No
P12	Poor	Weak	Unsuitable	No

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Example 1: Dining out data

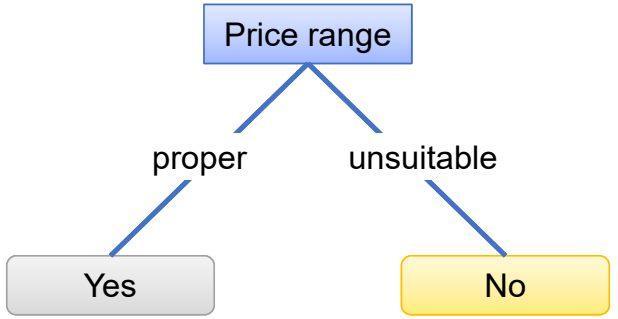



```
if Price_range='Proper' then Dine_out='Yes'
```


Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Example 2: Dining out (updated)

Restaurants	Food quality	Service speed	Price range	Dine Out?
P1	Good	Strong	Proper	Yes
P2	Good	Moderate	Proper	Yes
P3	Good	Moderate	Unsuitable	Yes
P4	Good	Weak	Unsuitable	No
P5	Average	Strong	Proper	Yes
P6	Average	Moderate	Proper	Yes
P7	Average	Moderate	Unsuitable	Yes
P8	Average	Weak	Unsuitable	No
P9	Poor	Strong	Proper	Yes
P10	Poor	Moderate	Unsuitable	No
P11	Poor	Weak	Proper	Yes

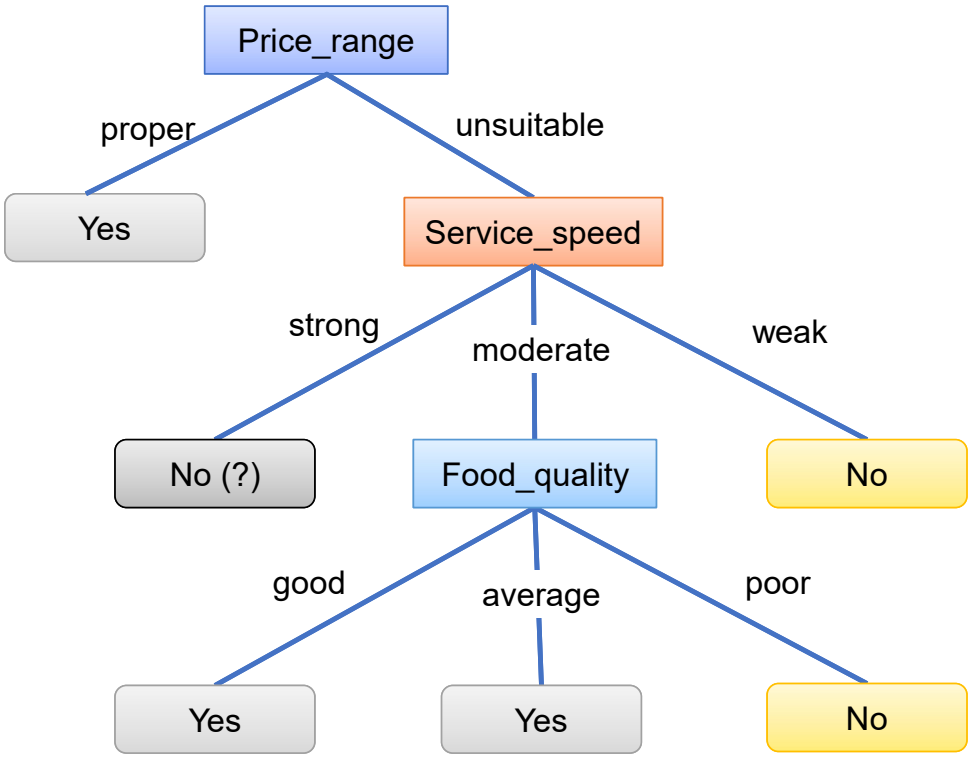


if Price_range='Proper'
then Dine_out='Yes'

?

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Example 2: Dining out (updated)



Restaurants	Food quality	Service speed	Price range	Dine Out?
P1	Good	Strong	Proper	Yes
P2	Good	Moderate	Proper	Yes
P3	Good	Moderate	Unsuitable	Yes
P4	Good	Weak	Unsuitable	No
P5	Average	Strong	Proper	Yes
P6	Average	Moderate	Proper	Yes
P7	Average	Moderate	Unsuitable	Yes
P8	Average	Weak	Unsuitable	No
P9	Poor	Strong	Proper	Yes
P10	Poor	Moderate	Unsuitable	No
P11	Poor	Weak	Proper	Yes

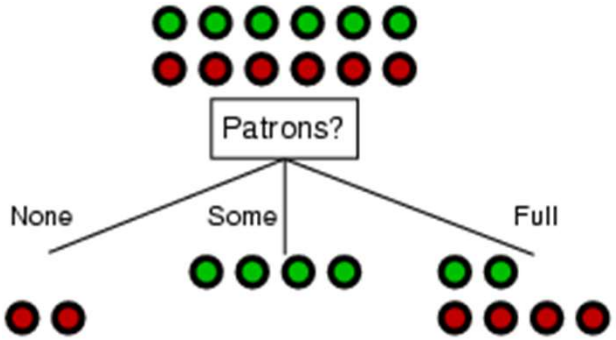
We need to build a Decision tree, but how do we decide which node (attribute) comes first?

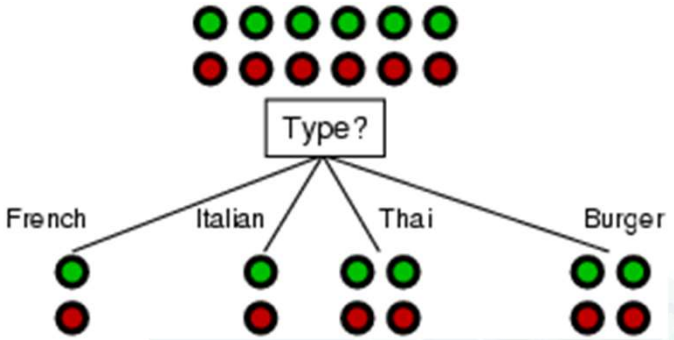
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Best Attribute

"Ideally" can classify the training examples into subsets, which has the same class

→ **Information Gain**





Information Gain

Nur ULFA Maulidevi

KK IF - Teknik Informatika- STEI ITB

Information Gain

Source: Machine Learning, Tom Mitchell

Pembelajaran Mesin
(Machine Learning)

Entropy (Information Content)

Measurement in Information Theory → impurity of an arbitrary collection of samples

S = set of training examples
Entropy: the minimum number of bits of information needed to encode the classification of an arbitrary member of S

A completely homogeneous sample has entropy of 0.

An equally divided sample has entropy of 1

$$Entropy(S) = \sum_{i=1}^{c} -p_i \log_2 p_i$$

- **S: set of training examples**
- **c: number of classes**
- **p_i: proportion of S belonging to class i**

Entropy (Information Content)

$$Entropy(S) = \sum_{i=1}^{c} -p_i \log_2 p_i$$

Data	Class(Y/N)
1	Y
2	Y
3	Y
4	Y

$$\begin{aligned} Entropy(S) &= -p_y \log_2 p_y \; - p_n \log_2 p_n \\ &= -\frac{4}{4}\log_2\left(\frac{4}{4}\right) - \frac{0}{4}\log_2\left(\frac{0}{4}\right) \\ &= 0 \end{aligned}$$

Data	Class(Y/N)
1	Y
2	N
3	Y
4	N

$$\begin{aligned} Entropy(S) &= -p_y \log_2 p_y \; - p_n \log_2 p_n \\ &= -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right) \\ &= 1 \end{aligned}$$

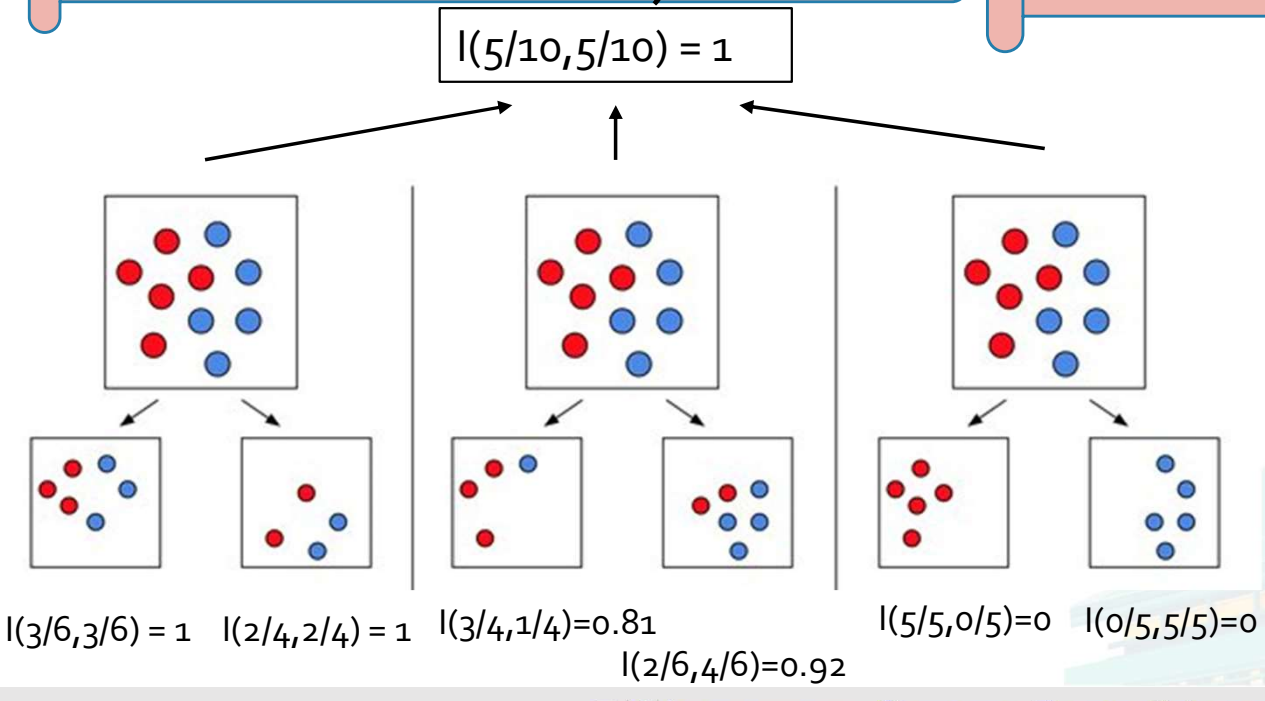
Entropy for S with 2 values/ classes

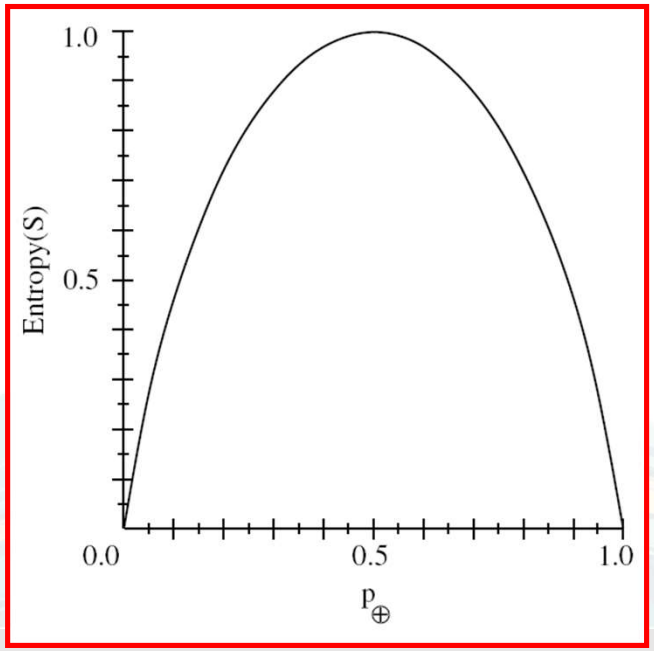
S contains positive examples and negative examples

$$Entropy(S) \equiv -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus}$$

Entropy 0: all examples belong to a class (no surprises, no message need be sent)

Entropy 1: $p_+ = p_-$
(1 bit is required to indicate the class of the drawn example)

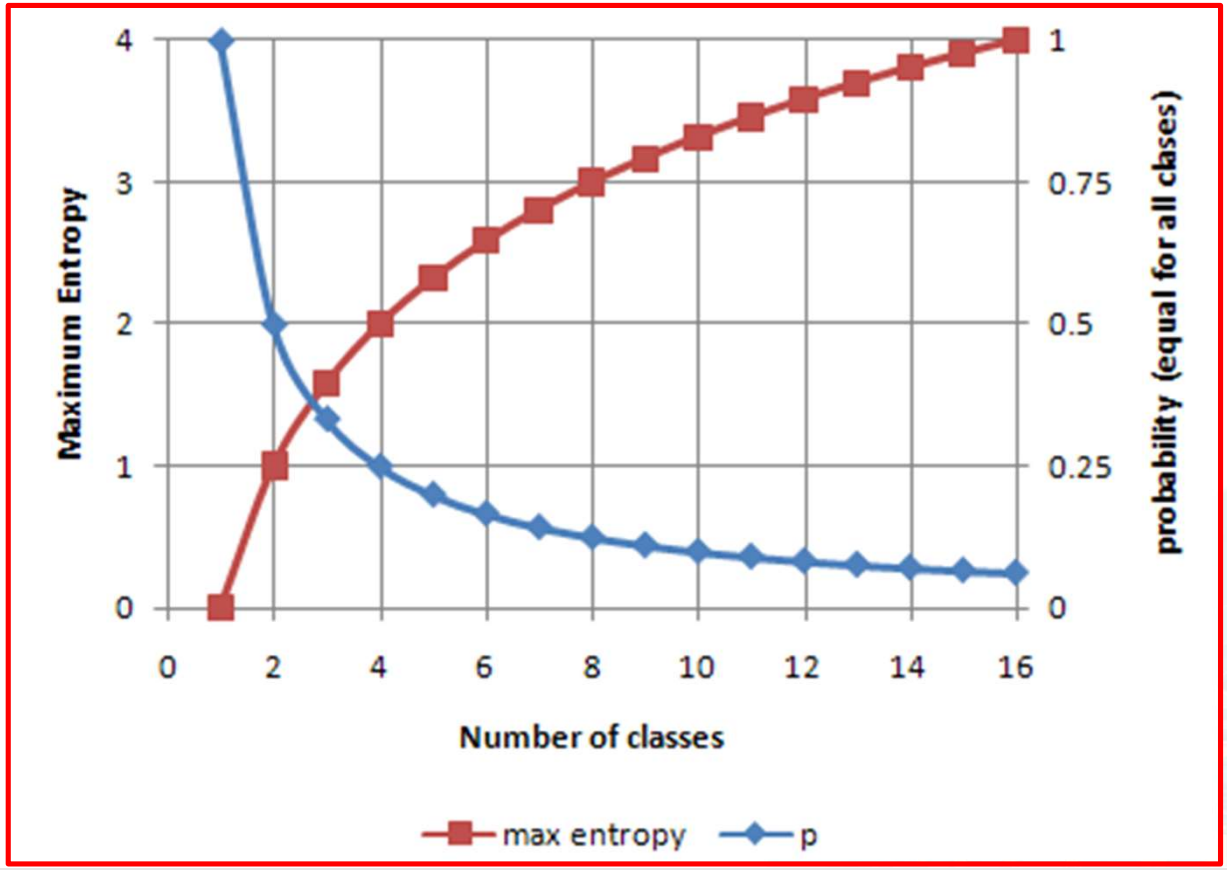




Entropy for S with n values/ classes

Target attribute has n values: entropy can be as large as

$$\log_2 n$$



Information Gain

Gain(S,A) = expected reduction of entropy due to sorting A
S: set of training examples; A: an attribute

Find A which has maximum Gain(S,A)

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

Example

Find the 'best' Attribute:
max(Gain(S,Outlook),
Gain(S,Temperature),
Gain(S,Humidity),
Gain(S,Wind))

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Entropy(S) = -\frac{9}{14}log_2\frac{9}{14} - \frac{5}{14}log_2\frac{5}{14} = 0.940$$

$$Gain(S, Outlook) = Entropy(S) - \sum_{v \in value\ of\ Outlook} \frac{S_v}{S} Entropy(S_v)$$

$$= 0.940 - [\frac{5}{14}Entropy(Sunny) + \frac{4}{14}Entropy(Overcast) + \frac{5}{14}Entropy(Rain)]$$

$$= 0.940 - [(\frac{5}{14}(-\frac{2}{5}log_2\frac{2}{5} - \frac{3}{5}log_2\frac{3}{5})) + (\frac{4}{14}(-\frac{4}{4}log_2\frac{4}{4} - \frac{0}{4}log_2\frac{0}{4})) + (\frac{5}{14}(-\frac{3}{5}log_2\frac{3}{5} - \frac{2}{5}log_2\frac{2}{5}))]$$

$$= 0.246$$

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Example

Find the 'best' Attribute:
max(Gain(S,Outlook),
Gain(S,Temperature),
Gain(S,Humidity),
Gain(S,Wind))

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Entropy(S) = -\frac{9}{14}log_2\frac{9}{14} - \frac{5}{14}log_2\frac{5}{14} = 0.940$$

$$Gain(S, Humidity) = Entropy(S) - \sum_{v \in value\ of\ Humidity} \frac{S_v}{S} Entropy(S_v)$$

$$= 0.940 - [\frac{7}{14}Entropy(High) + \frac{7}{14}Entropy(Normal)]$$

$$= 0.940 - [(\frac{7}{14}(-\frac{3}{7}log_2\frac{3}{7} - \frac{4}{7}log_2\frac{4}{7})) + (\frac{7}{14}(-\frac{6}{7}log_2\frac{6}{7} - \frac{1}{7}log_2\frac{1}{7}))]$$

$$= 0.151$$

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Example

Find the 'best' Attribute:
max(Gain(S,Outlook),
Gain(S,Temperature),
Gain(S,Humidity),
Gain(S,Wind))

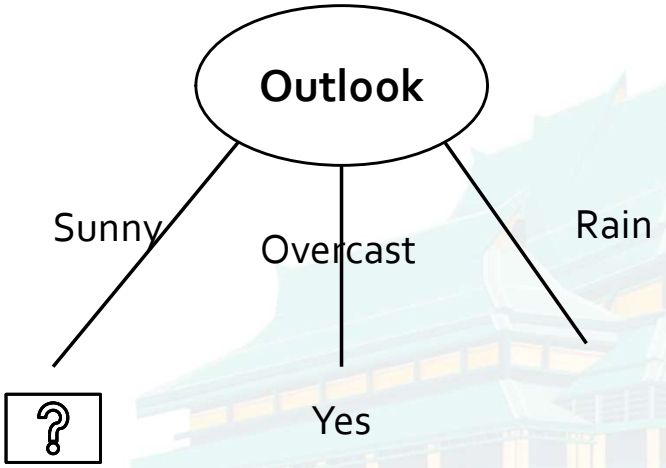
Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$Gain(S, Outlook) = 0.246$

$Gain(S, Humidity) = 0.151$

$Gain(S, Temperature) = 0.029$

$Gain(S, Wind) = 0.048$



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes

Find the 'best' Attribute:
max(Gain(S,Temperature),
Gain(S,Humidity),
Gain(S,Wind))

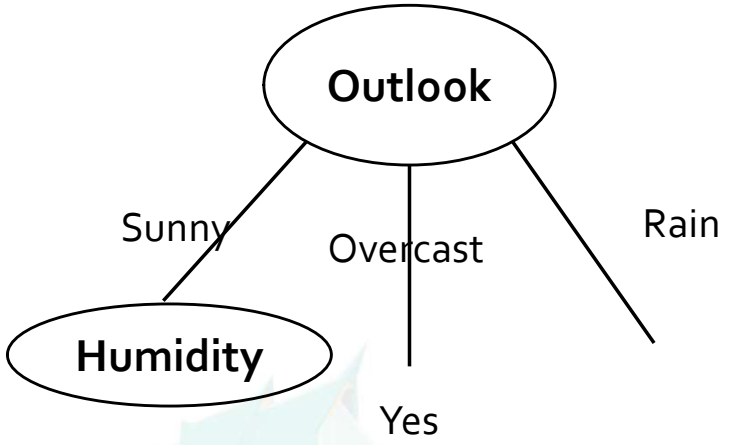
$$Entropy(S) = -\frac{2}{5}log_2\frac{2}{5} - \frac{3}{5}log_2\frac{3}{5} = 0.971$$

$$Gain(S, Humidity) = Entropy(S) - \sum_{v \in value\ of\ Humidity} \frac{S_v}{S} Entropy(S_v)$$

$$= 0.971 - \frac{3}{5}Entropy(High) - \frac{2}{5}Entropy(Normal)$$

$$= 0.971 - [(\frac{3}{5}(-\frac{0}{3}log_2\frac{0}{3} - \frac{3}{3}log_2\frac{3}{3}) + (\frac{2}{5}(-\frac{2}{2}log_2\frac{2}{2} - \frac{0}{2}log_2\frac{0}{2}))]$$

$$= 0.971$$



$$Gain(S, Humidity) = 0.971$$

$$Gain(S, Temperature) = 0.571$$

$$Gain(S, Wind) = 0.020$$

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Exercise (Class: Activity)

No	Deadline?	Ada Hangout?	Malas?	Aktivitas
1	Urgent	ya	Ya	Kumpul-kumpul
2	Urgent	Tidak	Ya	Belajar
3	Dekat	ya	Ya	Kumpul-kumpul
4	Tidak ada	Ya	Tidak	Kumpul-kumpul
5	Tidak ada	Tidak	Ya	Jalan-jalan ke mall
6	Tidak ada	Ya	Tidak	Kumpul-kumpul
7	Dekat	Tidak	Tidak	Belajar
8	Dekat	Tidak	Ya	Nonton TV
9	Dekat	Ya	Ya	Kumpul-kumpul
10	Urgent	Tidak	Tidak	Belajar

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial